

How to Learn Course Content Using AI

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Motivation

Why Use AI for Learning?

- Traditional learning: attend lecture → read notes → study alone
- **Problem:** When stuck, you have to wait for office hours or search online
- **AI-assisted learning:** Get instant, detailed explanations anytime

Key Idea

Use AI as your **personal course tutor** that has access to all your course materials.

What Can AI Help With?

Concept Understanding

- Explain each slide in detail
- Break down complex topics
- Provide additional examples

Coding Practice

- Generate executable code
- Add detailed comments
- Create variations to explore

Deep Dive & Verification

- Cross-check slide content
- Find missing details
- Clarify ambiguities

Self-Assessment

- Generate quizzes
- Test your understanding
- Identify weak areas

Tools

Tools You Need



- **Visual Studio Code:** Free code editor by Microsoft
- **Claude Code:** AI coding assistant by Anthropic
- Claude Code runs as an extension inside VS Code

Requirements

Software

- Visual Studio Code
- Node.js (version 18+)
- Claude Code (npm package)
- Claude Code VS Code extension

Account

- Claude.ai account
- Claude Pro or Max subscription
- Stable internet connection

Setup

Step 1: Install Visual Studio Code

1. Go to <https://code.visualstudio.com/>
2. Download for your OS (macOS / Windows)
3. Install:
 - **macOS:** Open .dmg, drag to Applications
 - **Windows:** Run .exe installer, check “Add to PATH”
4. Launch VS Code to verify

Step 2: Install Node.js

1. Go to <https://nodejs.org/>
2. Download the **LTS** version (v20+ recommended)
3. Install and verify:

```
$ node -version  
v20.x.x
```

Note

Node.js is required for installing Claude Code via npm.

Step 3: Install Claude Code

1. Open your terminal (Terminal on macOS, Command Prompt on Windows)
2. Install Claude Code globally:

```
$ npm install -g @anthropic-ai/claude-code
```

3. Authenticate by running `claude` (opens browser for sign-in)
4. Verify with `claude doctor` — all items should be green

Warning

Do **not** use `sudo` (macOS) or run as Administrator (Windows).

Step 4: Install Claude Code Extension in VS Code

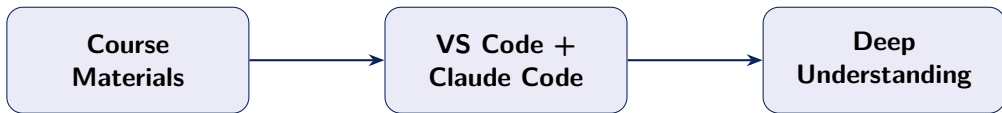
1. Open VS Code
2. Open Extensions panel:
 - macOS: `Cmd + Shift + X`
 - Windows: `Ctrl + Shift + X`
3. Search for “**Claude Code**”
4. Install the official extension by **Anthropic**
5. Sign in when prompted

Result

You now have VS Code with the Claude Code panel in the sidebar.

Learning Workflow

Overview: The Workflow



5 simple steps:

1. Create a folder and place all course materials
2. Open the folder in VS Code
3. Type `/init` in Claude Code
4. Set up Claude as your course tutor (with a prompt)
5. Start asking questions

Step 1: Prepare Your Course Folder

Create a folder and place **all** provided materials inside:

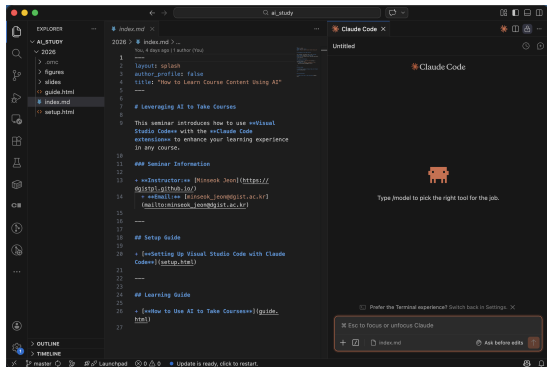
```
ProgrammingLanguages/  
  lec1.pdf  
  lec2.pdf  
  lec3.pdf  
  lec4.pdf  
  lec5.pdf  
  notes.pdf  
  ...
```

Tip

The more materials you provide (slides, notes, assignments), the better Claude can help.

Step 2: Open in VS Code

1. Launch Visual Studio Code
2. **File** → **Open Folder...** → Select your course folder
3. Open the Claude Code panel in the sidebar



Step 3: Initialize with /init

In the Claude Code panel, type:

```
This folder contains all the materials for my course. Read all  
the files including .pdf and .md files.
```

This creates a CLAUDE.md file that gives Claude context about your folder contents.

Step 4: Set Up Your AI Tutor

Type the following prompt in Claude Code:

Context:

Currently I am taking {Course Name} course
where all the contents are provided in the folder.

Role:

You are the instructor of the course who will
answer my questions about the course.

Command:

You need to answer my questions in detail
and step-by-step.

Format:

If I request, save the illustration as a .md file.

Replace {Course Name} with your actual course name.

Step 5: Start Asking Questions!

You are now ready to learn.

Claude can:

- Read your PDF slides and understand diagrams
- Provide detailed, step-by-step explanations
- Generate executable code examples
- Create quizzes to test your understanding
- Save explanations as .md files

Example Questions

Q1: Concept Check

Goal

Understand each slide of a lecture in detail.

Prompt:

```
For the first lecture (i.e., lec1.pdf),  
explain each slide in detail.
```

What Claude does:

- Reads the PDF file
- Explains each slide one by one
- Provides context and background for each concept

Q2: Coding Practice

Goal

Get executable code based on lecture content.

Prompt:

```
For the third lecture (i.e., lec3.pdf),  
I need you to implement programs that I can  
execute. I need you to add detailed comments  
illustrating the role of each line.
```

What Claude does:

- Creates runnable code files in your folder
- Adds line-by-line comments explaining each part
- You can run the code directly in VS Code's terminal

Q3: Deep Dive

Goal

Challenge slide content and verify correctness.

Prompt:

```
In page 15 of lec5.pdf, I guess the slide  
is wrong. The inference rule is missing some  
rules (the entire inference rules are  
provided in page 8).
```

What Claude does:

- Cross-references page 15 with page 8
- Identifies missing or incorrect rules
- Provides the corrected version

Q4: Self-Assessment

Goal

Test your understanding with quizzes.

Prompt:

```
Based on the lecture notes and slides for  
lecture 5, give me a quiz for verifying my  
understanding of the lecture.
```

What Claude does:

- Generates questions based on the lecture content
- Covers key concepts and potential exam topics
- Provides answers and explanations after you attempt

Tips

Tips for Effective Learning

1. **Be specific with references**

Always mention the exact file and page number (e.g., “page 15 of lec5.pdf”)

2. **Ask follow-up questions**

Don't stop at the first answer — ask “Why?”, “Can you give another example?”

3. **Save important explanations**

Ask Claude to save as .md files to build your own study notes

4. **Run the code yourself**

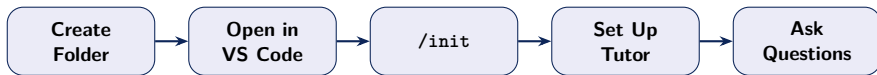
Execute generated code in VS Code's terminal, modify and experiment

5. **Quiz yourself regularly**

Use self-assessment before exams to identify weak areas

Summary

Summary



- Install: VS Code + Claude Code + Extension
- Place all course materials in one folder
- Use the structured prompt to set up your AI tutor
- Ask anything: concepts, code, deep dives, quizzes

Setup Guide: https://dgistpl.github.io/courses/ai_study/2026/

Thank you!

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